What a geometric tip-convolution simulator can and cannot predict: held-out validation against real AFM scans of calibration standards

First Author[a,*], Avish Cooray[a], Further Authors[a]

[a] *Department (TODO), Iowa State University, Ames, IA, USA*

Abstract

Atomic force microscopy (AFM) images are the sample topography dilated by the finite probe, and grey-scale morphological dilation is the standard geometric model of this effect. Simulators built on it are increasingly used to generate training data and benchmarks for tip deconvolution, yet they are rarely tested against real measurements under a protocol that separates what the data can validate from what they cannot. We present an open, browser-based and Python AFM forward simulator and validate it against 70 Bruker image files from ten folders recorded on two instruments (Dimension Icon and BioScope Resolve) on gratings, a step standard with a sphere probe, polymer films and fused silica. Each real frame is simulated on its own pixel grid with every candidate probe model. Sample geometry is calibrated on designated frames and transferred unchanged to held-out frames, and every reported claim was re-checked by three independent verification passes. The Python engine reproduces the browser page bit for bit over 73,410 configurations. On five grating folders, a measured rounded-square-pit model reproduces the held-out frames to 2.8–6.3 nm RMS (1.6–5.1 % of the 171–185 nm step), against 25–50 nm for the nominal sample presets, with 82–99 % of the misfit on the pit walls. The probe radius stated in the file header is not a usable simulator input: a 10 µm sphere is rejected by 52–80 nm step-edge widths, and an effective radius of at most about 150 nm follows only if the instrument's one-pixel fast-axis response is modelled. Conversely, none of these

*Corresponding author

Email address: alloydas@iastate.edu (First Author)

samples identifies the tip at the nominal radius scale ($R \lesssim 12$ nm), the yaw sign, or the absolute XY/Z calibration. We report these limits explicitly as the boundary conditions for inverse-problem benchmarks built on such simulators.

Keywords: atomic force microscopy, tip convolution, morphological dilation, forward model, validation, calibration gratings, blind tip reconstruction

---

1. Introduction

The atomic force microscope [1] records topography by scanning a sharp probe over a surface. Because the probe has a finite size and shape, the image is not the surface: narrow trenches are inaccessible, sharp features are broadened and rounded, and damaged or multi-apex probes produce duplicated features. In the height-field approximation this tip–sample interaction is grey-scale morphological dilation of the surface by the reflected tip, and erosion with a known tip gives the tightest surface consistent with the image [2, 3]. Villarrubia's algorithms for image simulation, surface reconstruction, blind tip reconstruction (BTR) and certainty maps [3] are implemented in standard analysis software such as Gwyddion [4, 5].

Forward simulators built on this model have gained a second role: generating paired true/measured data for learned restoration and tip estimation. Convolutional networks trained on synthetic distorted/ground-truth pairs correct AFM artefacts [6] and recover nanoscale profiles [7]; differentiable dilation and erosion make BTR robust to noise [8]; synthetic generators with tip convolution, noise, debris and feedback artefacts train models for autonomous AFM [9]; and multi-view neural reconstruction models commercial pyramidal tips parametrically [10]. Independent tip characterisation by electron microscopy, critical-dimension AFM or reference trenches [11, 12] provides the probe ground truth these methods need.

All of these uses assume that the simulator reproduces real images when it is given the right sample and probe. That assumption is seldom tested directly against real scans, and when it is, the test rarely states what the data could not have revealed. A pixel-level agreement on a grating, for example, may say nothing about the tip if the grating walls are shallower than every probe facet. This distinction matters for inverse-problem benchmarks: a simulator validated only where the tip is invisible cannot certify tip-dependent

conclusions.

This paper reports such a validation. Its contributions are:

1. an open forward simulator (interactive 2D and 3D browser applications and a vectorised Python engine) whose engine is shown to reproduce the browser page bit for bit (Section 2);
2. a validation protocol for real Bruker data with a case catalogue, explicit calibrate/predict roles, candidate-probe sweeps, a no-tip control and an evidence tag on every number (Section 3);
3. held-out results on ten folders of real data, with each headline claim re-checked by three independent verification passes (Section 4); and
4. an explicit list of what these measurements validate and what they cannot, which sets the boundary conditions for the inverse problem (Section 5).

2. Forward model and implementation

2.1. Imaging model

A surface is a height field $S(\mathbf{x})$ over lateral positions $\mathbf{x} = (x, y)$, and a probe is represented by its lower height envelope $T(\mathbf{d}) \geq 0$, the height of the tip above its apex at lateral offset $\mathbf{d}$, with $T(\mathbf{0}) = 0$. Contact-mode imaging is computed as dilation,

$$I(\mathbf{x}) = \max_{\mathbf{d}} \big[\, S(\mathbf{x} + \mathbf{d}) - T(\mathbf{d}) \,\big], \tag{1}$$

and the classical reconstruction with a known tip is the erosion

$$E(\mathbf{x}) = \min_{\mathbf{d}} \big[\, I(\mathbf{x} - \mathbf{d}) + T(\mathbf{d}) \,\big], \tag{2}$$

which is an upper bound on S and equals it only where the tip touched the surface at a single point [3]. Forward imaging always uses Eq. (1), also for the concave probe model; erosion is available only as an explicit inverse option.

The model is geometric. It does not represent cantilever dynamics, feedback control, adhesion, friction, sample or tip deformation, or overhanging surfaces. These boundaries are deliberate and are part of what Section 4 tests. Figure 1 illustrates the model on a synthetic profile.

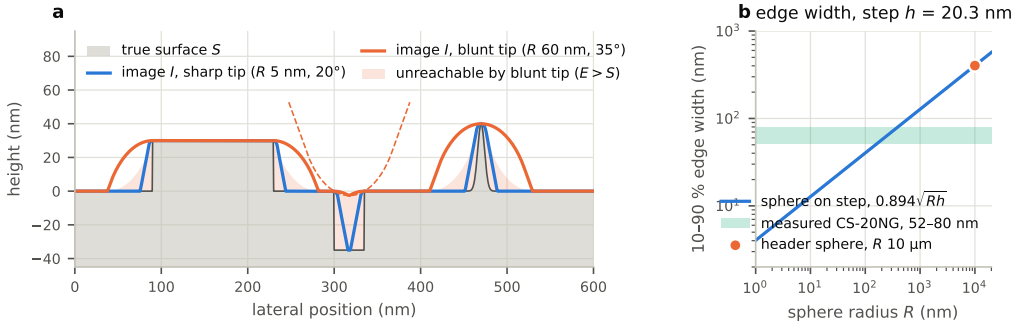


Figure 1: The imaging model. (a) A synthetic profile (grey) imaged by Eq. (1) with a sharp sphere–cone tip ($R = 5$ nm, 20° half-angle, blue) and a blunt one ($R = 60$ nm, 35°, orange; dashed: the tip at the trench centre). The blunt tip cannot enter the 35 nm trench and broadens the asperity. Shaded: where the erosion E of the blunt-tip image (Eq. (2)) lies above the true surface, i.e. surface the tip never reached. (b) 10–90 % edge width of a sphere on a 20.3 nm step, Eq. (4), against the radius. The edge widths measured on CS-20NG (52–80 nm, Section 4.2) are 5–8 times narrower than the 403 nm the header's 10 µm sphere implies. The closed form ignores the instrument's pixel response, so it does not by itself give the radius.

2.2. Probe and surface libraries

The analytical probe library contains conical, spherical, hyperboloidal, flat-punch, concave (inverse sphere), asymmetric, double-tip and combined sphere–cone tips. The 3D application adds a faceted pyramid with unequal front, back and side angles and a rounded apex, with default dimensions taken from a commercial probe specification. Probe meshes for the probes used in this study (RTESPA, ScanAsyst-Air, SNL-10 and SAA-SPH) are generated as Wavefront OBJ files. An independent tool checks them by casting vertical rays against the triangles and comparing the resulting lower envelope with the analytical tip equation. The surface library ranges from periodic waves, particles, pits and trenches to rough, chirped and lattice surfaces, plus models of the calibration samples used here (Section 3.1). Imported OBJ meshes are rasterised to their upper envelope (surfaces) or lower envelope (probes).

2.3. Implementations

Browser applications. A 2D line-scan simulator and a 3D raster simulator are written as static HTML/JavaScript (Three.js rendering). The 3D page computes an 80×80 height map and exports true and measured maps (CSV

and 16-bit PNG), parameters, metadata and OBJ meshes. The animated probe is only a visualisation layer: the measured map is always computed from the full discrete dilation.

Python engine. `afm_forward_sim` has two layers. The *parity layer* reproduces the page's computation on its 80×80 grid, including its kernel truncation and height clamp. The *physical layer*, used for every real-scan simulation in this paper, applies the same tip and surface geometry on arbitrary grids. It dilates on a fine grid of spacing Δ/u (pixel size Δ, upsampling $u \geq 8$ in full runs) and point-samples at pixel centres. Vertical grating walls are dilated exactly from their outlines; sloped walls go through the fine lattice, whose error falls roughly as $1/u$. For apex radii beyond the page's lookup table ($R > 12$ nm), pyramids use the exact morphological opening by the apex ball.

Parity. A test harness drives the 3D page in headless Chrome: it clicks the same buttons and moves the same sliders as a user, pins the random seed and reads back every intermediate array. Over 73,410 configurations of tip, surface, scan size, slider extremes, toggles and noise, the parity layer matched the page with zero mismatches and a maximum absolute difference of 0 nm [SIM]. Parity shows that the page and the engine model the same geometry. It does not show that the page's 80×80 grid can reproduce real scans; that is the physical layer's role.

3. Validation data and methods

3.1. Data

Table 1 lists the ten data folders. They were recorded in PeakForce Tapping mode on a Bruker Dimension Icon and a Bruker BioScope Resolve, on two square-pit gratings (VGRP-15M, VGRP-GS), a step standard (CS-20NG), a three-dimensional grating (TGXYZ02), two polymer films (PSFILM-12M, PS-LDPE-12M) and fused silica (FSILICA-12M). Of the 70 image files, 51 were simulated and compared. The case catalogue excluded 19, each with a stated reason (for example feedback oscillation, drift shear or a wrong header scan size). A machine-readable case catalogue records, for every folder, the instrument, decoding conventions, nominal and measured sample models, candidate probe models, the role of every scan and the known artefacts.

Table 1: Data folders. Probe: folder label and file-header values, which were not independently checked. Frames: compared / catalogue files. Class: analysis driver.

Sample	Instr.	Probe label	Header probe	Frames	Class
FSILICA-12M	Icon	ScanAsyst Air Probe2	ScanAsyst-Air, R 5 nm	5/11	flat reference
PSFILM-12M	Icon	ScanAsyst Air Probe2	ScanAsyst-Air, R 5 nm	17/17	polymer
PS-LDPE-12M	Resolve	ScanAsystAir	R 5 nm	8/8	polymer
VGRP-15M	Icon	RTESPA 300 Probe2	RTESPA-300-125, R 125 nm	4/5	grating
VGRP-15M	Icon	ScanAsyst Air Probe4	ScanAsyst-Air, R 5 nm	6/6	grating
VGRP-GS	Resolve	RTESPA-150	R 8 nm	2/10	grating
VGRP-GS	Resolve	ScanAsystAir	R 5 nm	3/4	grating
VGRP-15M	Resolve	SNL10 B	R 8 nm, k 42 N/m	3/4	grating
CS-20NG	Icon	SAA_SPH_10UM Probe4	SAA-SPH-10UM, R 10 µm	3/3	sphere edges
TGXYZ02	Icon	SAA_SPH_10UM Probe4	SAA-SPH-10UM, R 10 µm	0/2	negative control
Total				51/70	

3.2. Decoding, masks and conventions

Height channels are decoded following Gwyddion's NanoScope importer: height = raw × HV / 2^{8B} × S, with raw the signed little-endian integer at the stored width, HV the hard value of the Z scale, B the declared bytes per pixel and S the sensitivity. The reader is bit-identical to reference decoders on all 136 files tested [MEAS]. The metrology channel is the Height Sensor, because the Height (piezo drive) channel reads 0.4–2.5 % high and carries creep and hysteresis [MEAS]. Arrays are kept in stored order and pixel (i, j) is centred at $(x, y) = (j\Delta_x, i\Delta_y)$ with $\Delta = L/N$ for scan size L and N pixels.

No metric uses an invalid pixel. Masks combine zero-filled dropout runs, catalogue-listed corrupted segments, and rows rejected by the registration; railed channels are flagged and never read as topography.

3.3. Levelling, registration and sample models

Images are levelled with robust plane and per-line fits on a reference-level mask; for gratings a two-level segmentation (Otsu thresholding [13] with morphology) defines that mask. A square lattice is registered in three steps: an FFT estimate, least squares on feature centroids, and a model-image fit of rounded rectangles with a linear wall and Gaussian blur. The result is the full lattice affine (per-axis scale and shear), which places the sample model in image coordinates. In synthetic round trips this registration leaves a model floor of small wall and corner deltas and a 0.3–0.9 nm depth bias [SIM].

Each grating has three sample models:

M0 the simulator's built-in preset: 5 µm sharp, vertical-walled pits;

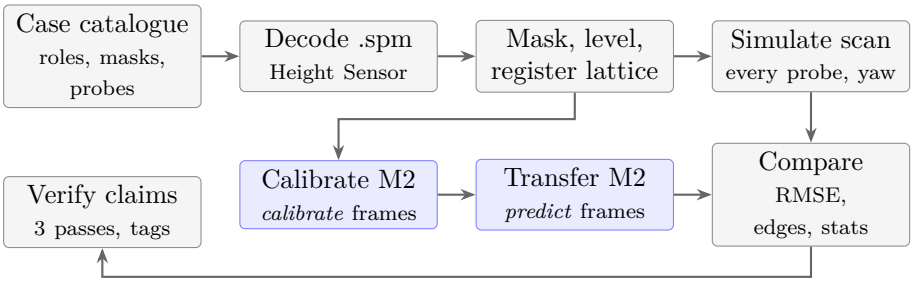


Figure 2: Validation pipeline. The case catalogue fixes every scan's role before any fit. The measured sample model M2 is calibrated through the simulator on *calibrate* frames only and transferred unchanged to *predict* frames, where only the registration and levelling are refitted. Each frame is simulated with every candidate probe, both yaw signs and a no-tip control. Every reported claim then carries an evidence tag and passes three verification lenses.

M1 an earlier hand-built model of 6 µm wells;

M2 a measured rounded-square-pit model with six parameters (depth, two half-widths, corner radius, wall run and wall blur), calibrated through the simulator on the folder's *calibrate* frames only.

On each *predict* frame, M2 is transferred unchanged and only that frame's registration and levelling are refitted; this is the held-out comparison $\mathrm{M2}_{\mathrm{holdout}}$. The calibration is probe-consistent: it is repeated for every probe hypothesis, so the geometry is the one that, dilated by that probe, best matches the calibrate frames. A cross-probe pair (same sample, instrument and mounting, two probes) fits the geometry on one probe's frames and predicts the other's. Figure 2 summarises the pipeline.

3.4. Probe hypotheses, orientation and sampling

Every probe hypothesis in the catalogue is simulated: header values, folder-label values, radius sweeps (2–250 nm) and a no-tip control, which isolates the tip's contribution. The engine's tip frame has the cantilever axis across the fast scan at yaw 0°; at Bruker scan angle 0 the fast axis runs along the cantilever, so the physical geometry is yaw ±90° [INF]. The sign is not established, so both signs and the yaw-0° control are simulated and reported; a yaw selected by a residual is a fit, not a measurement.

The simulation point-samples each pixel, while the instrument averages 2.0–15.1 PeakForce cycles per pixel along the fast axis [MEAS]. Where a

driver models this fast-axis pixel response, it says so, and Section 4.2 shows where the choice matters.

3.5. Metrics

For registered samples the residual is $r = I_{\mathrm{sim}} - I_{\mathrm{meas}}$ on valid pixels, with no fitted offset, and

$$\mathrm{RMSE} = \left(\frac{1}{N} \sum r^2 \right)^{1/2}. \tag{3}$$

It is split into a wall band (pixels within the wall extent plus two pixels of the registered outline) and the remaining flat pixels. Tip discrimination within a folder is the largest image difference between any two probe hypotheses, each after its probe-consistent calibration, divided by the scan's residual floor. A value below 1 means that the hypotheses cannot be told apart. For the step standard we use 10–90 % edge widths and joint fits of geometry, fast-axis response and sphere radius. A sphere of radius R imaging a vertical step of height h gives a 10–90 % edge width

$$w_{10\text{–}90} = \left(\sqrt{1.8} - \sqrt{0.2}\right)\sqrt{Rh} \approx 0.894\sqrt{Rh}, \tag{4}$$

and walls steeper than about 4° cannot make it narrower. The random-surface presets of the polymer and fused-silica samples are not registered to the real topography, so those samples are compared statistically (background S_q, autocorrelation length, power spectral density and feature statistics). They are also compared in a cross-scale test: a measured finer zoom is resampled through the engine onto its parent frame.

3.6. Evidence tags and verification

Every reported number carries a basis tag: [MEAS] (derived from the measured images alone), [SIM] (from the simulator, alone or compared with a measurement) or [INF] (a conclusion drawn from both). Each run records checksums of its inputs (engine, modules, catalogue, driver) so that results from older code are flagged, never presented as current. The four main claims of each folder (40 in total) were then checked by three independent verification passes: *reproduce* (re-derive the numbers from the outputs and from the raw files with independent code), *method* (circularity, conventions, uncertainty) and *scope* (wording strength, alternative explanations). None of the 40 claims was refuted. All 40 required qualification of scope, uncertainty or wording, and the statements below use the qualified versions. A separate report recomputed 22,596 reported values from the saved maps; all matched within 0.01 nm.

Table 2: Grating results with the folder-label probe at yaw +90° [SIM]. Held-out: number of predict frames (independent sites). RMSE: mean (range) over held-out frames. Discr.: tip discrimination ratio (< 1: probe hypotheses indistinguishable). Depth: measured step, mean ± SE over frames [MEAS].

Folder	Held-out	$M2_{holdout}$ (nm)	M0 (nm)	M1 (nm)	Discr.	Depth (nm)
VGRP-15M Icon, ScanAsyst-Air	3 (3)	3.45 (3.15–4.03)	49.6	25.9	0.089	184.54 ± 0.06
VGRP-15M Icon, RTESPA-300	3 (2)	4.91 (4.67–5.08)	49.5	25.8	0.61	184.07 ± 0.06
VGRP-GS Resolve, RTESPA-150	1 (1)	2.79	31.0	49.2	0.13	171.04 ± 0.09
VGRP-GS Resolve, ScanAsyst-Air	1[a]	3.93[b]	32.6	45.5	0.12	171.94 ± 0.06
VGRP-15M Resolve, SNL10 B	2[c]	6.34 (3.84–8.84)	47.5	31.0	0.16	174.68 ± 0.02

[a] Weak hold-out: same site as a calibrate frame, probably the same pits. [b] 3.59 nm with PeakForce pixel averaging. [c] One field imaged twice.

4. Results

4.1. Gratings: the measured sample model transfers to held-out frames

Table 2 and Fig. 3 summarise the five grating folders. With the folder-label probe, $M2_{holdout}$ reproduces the held-out frames to 2.8–6.3 nm RMS per folder (single frames 2.8–8.8 nm, 1.6–5.1 % of the 171–185 nm step) [SIM]. The nominal preset M0 leaves 31–50 nm and M1 25–49 nm, 5–18 times more. These baselines show that the nominal pit size, vertical walls and sharp corners are wrong; they do not measure predictive skill.

The Icon VGRP-15M folder with the ScanAsyst-Air probe is the strongest case. Its calibrate frames come from one stage site, and its three held-out frames from three non-overlapping sites 0.65–1.07 mm away. The probe-consistent calibration gives half-depth widths of 5770×5793 nm, 1187 nm corner radii, a 184.60 nm depth and a 152 nm 10–90 % wall. Transferred unchanged, it reproduces the held-out frames at 4.03, 3.16 and 3.15 nm (mean 3.45 ± 0.29 nm, SE over sites). Calibrating the same model on each held-out frame itself improves this by only 0.05–0.36 nm, and leave-one-site-out calibration gives site means of 3.47–3.67 nm [SIM].

The residual sits on the walls. The wall band holds 12–18 % of the pixels but 82–87 % of the squared residual (band RMSE 6.8–10.9 nm). The flat-region RMSE (mean 1.455 nm) equals the frames' own flat-level scatter (1.454 nm). That equality holds for any flat model with the correct depth, so it confirms only the transferred depth, not the lateral geometry [INF]. Figure 4 shows the same pattern on every held-out grating frame.

The remaining wall misfit comes from effects the model omits, not from the tip:

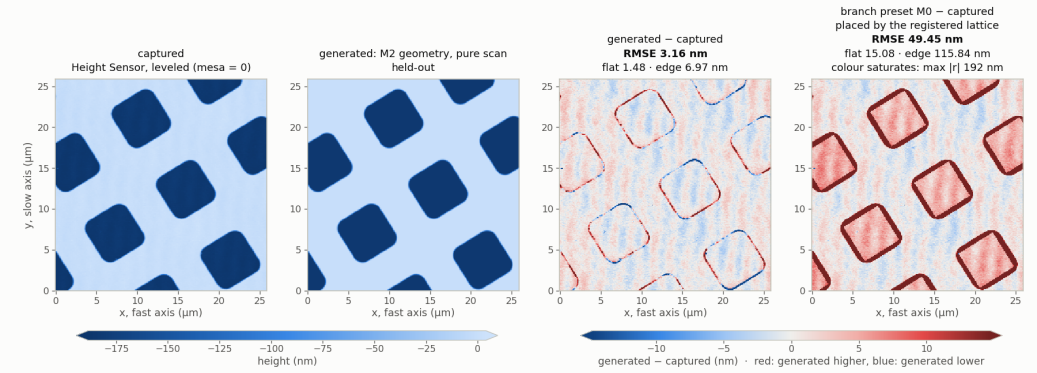


Figure 3: Held-out prediction on VGRP-15M (Dimension Icon, ScanAsyst-Air probe, frame 00004, 256 × 256 pixels at 101.3 nm/pixel). From left: measured Height Sensor image (levelled, mesa = 0); the pure simulated scan of the M2 geometry calibrated on other frames at another site; simulated minus measured (RMSE 3.16 nm; flat 1.48 nm, wall 6.97 nm); the same for the nominal preset M0 placed by the registered lattice (RMSE 49.45 nm). The difference colour scale is clipped at 14.7 nm.

- open-loop fast-axis scanner non-linearity, about 0.4 % per 10 µm (RTESPA-150);
- drift shear (VGRP-GS);
- PeakForce feedback on fast-axis walls (SNL10 B; 15.5 nm on its fast calibrate frame);
- site- and session-dependent pit widths (the RTESPA-300 held-out pits image 2–55 nm narrower; cause not identified).

The tip is not identifiable on the gratings. Every probe hypothesis, the no-tip control included, gives the same held-out RMSE to within 0.05 nm on three of the five folders (for example 3.447 nm without a tip and 3.459 nm with $R = 12$ nm). The discrimination ratio is below 1 in every folder (0.089–0.61). The reason is geometric: the 43–49° pit walls are shallower than every probe facet, and the probe-consistent calibration absorbs the apex into the pit width. With the geometry held fixed, dilation by $R = 125$–250 nm is visible (up to 19.4 nm RMS) but disappears after recalibration (1.6 nm). The raw fixed-geometry difference is therefore not a discrimination test [SIM]. Figure 5 shows this on the two Icon VGRP-15M folders. With the nominal geometry held fixed, a larger apex raises the held-out RMSE by up to 10.8 nm

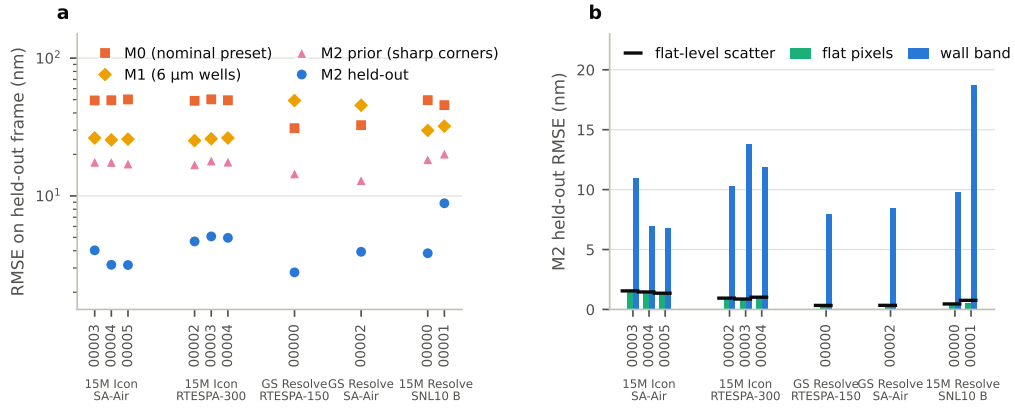


Figure 4: All ten held-out grating frames (label probe, yaw +90°), grouped by folder in the order of Table 2. (a) RMSE of the three sample models (log scale): the transferred measured model M2 against the nominal preset M0, the 6 µm wells M1 and the catalogue prior with sharp corners. (b) M2 held-out RMSE on the flat pixels and in the wall band, with each frame's own flat-level scatter. The flat error is at or below the scatter on every frame; the misfit is on the walls.

($R = 250$ nm). After the probe-consistent recalibration, the same probes change it by at most 0.52 nm on the RTESPA-300 folder and 0.011 nm on the ScanAsyst-Air folder.

Cross-probe pair. Only the Icon VGRP-15M pair tests the geometry that each probe hypothesis implies. Relative to a ScanAsyst-Air apex near its nominal radius, the header's 125 nm apex for the RTESPA-300 probe is inconsistent with the pit widths in a post-hoc site-level two-sample test (9.4 SE, $p = 2.3 \times 10^{-4}$), while radii up to about 30 nm are consistent [INF]. The pre-registered decision rule returns only "mixed" for this hypothesis, because with three or four target sites its pooled-t critical values (19.2 and 9.22) are out of reach. The test therefore lacks power; it is not conflicting evidence. Neither the pair nor the folder singles out $R = 8$ nm.

Instrument readings. Within a session the step depth repeats to an SE of 0.02–0.09 nm over frames [MEAS]. Between probes on the same die the depth differs by 0.47 nm (Icon) and 0.95 nm (Resolve). The Resolve reads the VGRP-15M depth at 0.947–0.949 of the Icon's, although it is unknown whether the two instruments imaged the same die [MEAS]. These are session values, not traceable calibrations: the 180 nm nominal depth has no stated

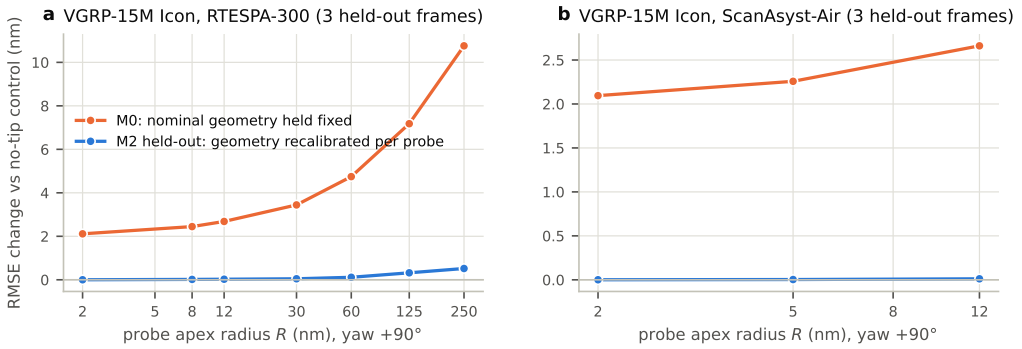


Figure 5: Why the gratings do not identify the tip. Change of the held-out RMSE relative to the no-tip control, mean over the three held-out frames, against the simulated apex radius: (a) VGRP-15M Icon, RTESPA-300 folder (no-tip control: M0 47.09 nm, M2 4.89 nm); (b) VGRP-15M Icon, ScanAsyst-Air folder (M0 47.52 nm, M2 3.448 nm). Orange: nominal geometry held fixed, so the dilation is visible. Blue: the measured geometry recalibrated for each probe absorbs the apex into the pit width. The ranking along the blue line is not tip evidence.

tolerance, and every pitch was assumed exact.

4.2. Step standard: header tip metadata is not a valid simulator input

The CS-20NG frames were recorded with a probe whose header names a 10 µm sphere (SAA-SPH-10UM). For the measured 19.9–20.4 nm steps, Eq. (4) and the engine both predict 10–90 % edges of about 400 nm [SIM]. The measured edges are 52–80 nm, or 42–117 nm across the estimators and edge families tried, which is 3.4–10 times narrower and close to the one-pixel sampling floor [MEAS](Fig. 6). Pixel averaging, feedback lag, drift, calibration, yaw or holder tilt can only widen the predicted edge, so none of them closes the gap. Joint fits that refit geometry, wall and fast-axis response at each radius reject $R = 10$ µm in every variant: the pooled paired t is 21.8–22.0 over 309 edge segments, and 3.55 against a critical 2.92 with the frame as the statistical unit [INF].

The effective radius is bounded only conditionally. With the instrument's fast-axis response modelled (a box about one pixel wide, consistent with averaging over the 3.91 ms dwell of 7.8 PeakForce cycles), the three frames give a conservative upper bound of about 150 nm on the sphere-equivalent apex radius over the bottom 20 nm of the probe. The bracket is 130–153 nm across accepted variants; the best fits are 76–95 nm, and there is no lower

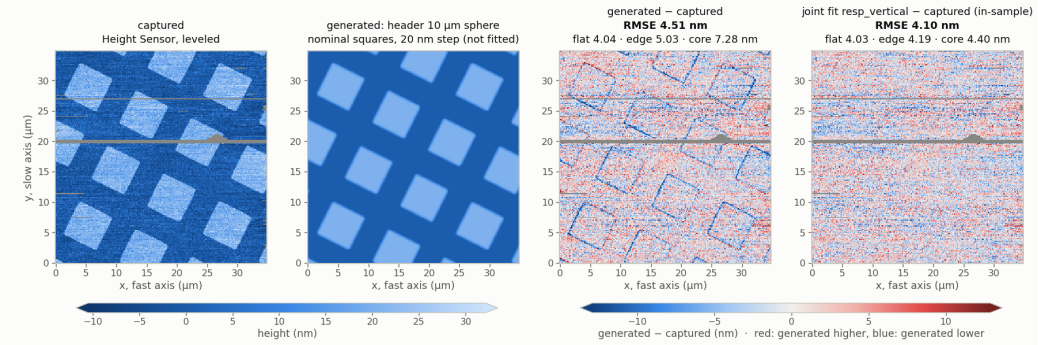


Figure 6: CS-20NG step standard (Icon, frame 00000, 136.3 nm/pixel). From left: measured image; the forward prediction with the header's 10 µm sphere on the nominal squares (placement registered, nothing else fitted); its residual (RMSE 4.51 nm); the residual of the in-sample joint fit of geometry, fast-axis response and a sphere radius pooled over three frames (RMSE 4.10 nm). Grey: masked pixels of the captured frame (dropout rows, particles), not compared; the generated map is the pure, unmasked scan. The 10 µm sphere is rejected by the edge widths, not by these image RMSEs: its 0.9 µm broadening partly offsets the nominal-width error.

bound [INF]. The simulator's own pixel-centre sampling is rejected (paired $t = 8.9$): it cannot reproduce the measured fast/slow edge anisotropy (for example 117/65 nm), and with it the bound would be 476 nm. The tip-radius conclusion therefore depends on modelling the pixel response, while the rejection of the header radius does not. The data cannot distinguish a sharp probe from a 10 µm sphere carrying an asperity or debris particle protruding about 20 nm, which would give 24–59 nm edges [SIM].

Nominal-geometry image RMSEs rank the probes wrongly: they favour the 10 µm sphere. Only geometry-refitted fits rank probes correctly. Figure 7 shows the edge evidence directly.

4.3. Polymers and fused silica: statistics and cross-scale consistency

The random-surface presets for these samples are not registered to the real topography, so only statistics can be compared, and each preset fails in a specific way [INF]:

- the fused-silica preset (S_q roughness 0.3 nm) is 26–54 % too smooth (ratio 0.46–0.74, falling with frame size) and has no particles;
- the PSFILM preset is about 3.5 times too rough in the background, yet lacks the nodules that carry 96 % of the measured variance;

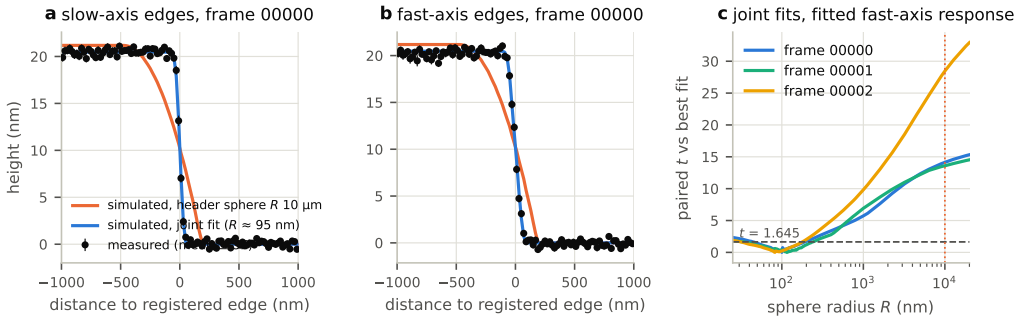


Figure 7: CS-20NG edge evidence. (a, b) Mean edge profiles of frame 00000 against the distance to the registered outline, for edges crossing the slow and the fast scan axis: measured (mean ± SE per 20 nm bin), the joint fit with the header's 10 µm sphere, and the best joint fit (fitted fast-axis response; $R \approx 95$ nm pooled). (c) Paired segment t of each radius against the best fit, per frame, for the same model. The header radius (dotted) is rejected on every frame (t 13.6–28.5). Single frames bound the radius at 193, 248 and 180 nm; the pooled bound is 153 nm, and it holds only with the fast-axis response modelled.

- the PS-LDPE preset cannot represent the domain coverage (0.33 against 0.135–0.199) or shape (aspect ratio 2.2 against 1.0).

A fused-silica fit on the calibrate frames matched a held-out frame's S_q (ratio 0.989). Copying the calibrate frames' median S_q does as well (0.987), so this is not predictive skill.

A different test uses only measured data. Each finer zoom, resampled through the engine onto its parent frame (Fig. 8), reproduces the parent to 0.28–0.50 nm (FSILICA), 0.20–0.72 nm (PSFILM) and 1.0–1.6 nm (PS-LDPE) RMS [INF]. This checks the decoding, including per-file auto-ranged Z scales (consistent to 2–3 %), the pixel conventions, the registration and the resampling layer. It does not depend on the tip, because dilation changes these images by less than the noise; on PS-LDPE the dilation is exactly zero at the recorded pixel sizes. On the soft LDPE phase, part of the residual reflects the different PeakForce setpoints of zoom and parent, which a geometric model cannot represent. Figure 9 summarises both comparisons.

Robust blind tip reconstruction [3] on the tip-sensitive zooms excludes only blunt apexes. On PSFILM an isotropic apex of 120 nm or more is excluded, but a blunt apex extended only along x is not. Every sharp candidate ($R \leq 12$ nm) passes, which demonstrates no power, and every radius bound depends on the reconstruction's threshold settings [INF].

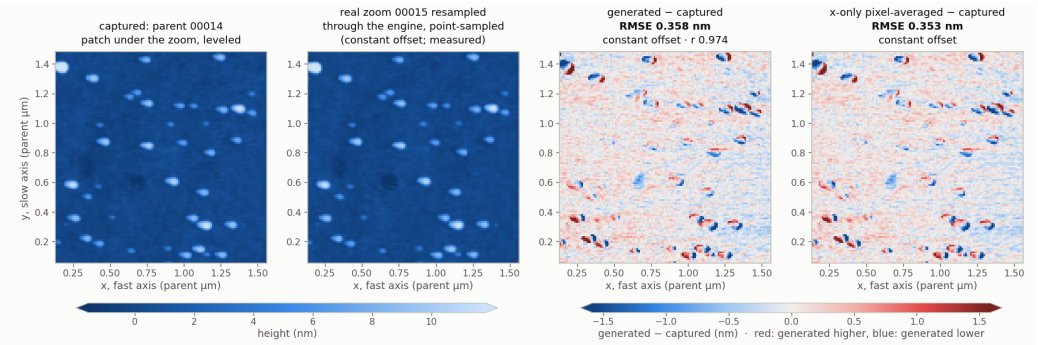


Figure 8: Cross-scale test on PSFILM (Icon, ScanAsyst-Air). From left: parent frame 00014 under the zoom (11.36 nm/pixel); the measured zoom 00015 (5.81 nm/pixel) resampled through the engine onto the parent grid; their difference (RMSE 0.358 nm, $r = 0.974$); the same with fast-axis-only pixel averaging (0.353 nm). The test is measured against measured and does not depend on the tip at these pixel sizes.

4.4. Negative control: the pipeline reports failed tracking instead of fitting it

The two TGXYZ02 frames hold no tracked topography [MEAS]. In frame 00000 the Z drive stayed at its retract limit for the whole frame: every one of the 65,536 Height pixels holds the same limit value, and every captured force curve lies within 0.95 nm of one template. In frame 00001 the peak-force error is below $-$setpoint/2 in 84.7 % of pixels. The pipeline flags both frames instead of fitting them. For the idealised model its sphere-on-step images equal the closed form to 0 nm, but neither frame matches any prediction, and this mismatch says nothing about the simulator.

5. Discussion

5.1. What was validated

Within its geometric assumptions, the simulator, given a measured sample model and a probe of roughly the right size, reproduces real grating images at held-out sites to a few nanometres. The remaining misfit lies on the walls, and identifiable instrument effects explain much of it. The engine is identical to the browser page, its dilation equals closed forms where they exist, and its decoding, pixel conventions and resampling are consistent across scales on real data.

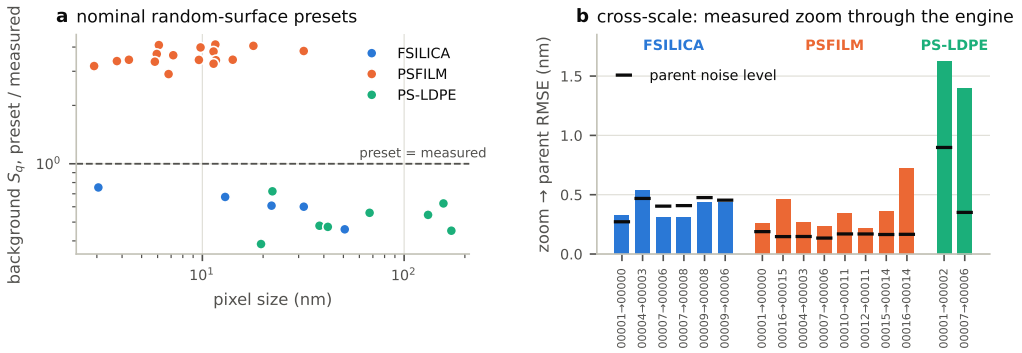


Figure 9: Polymers and fused silica. (a) Background S_q of the nominal random-surface preset, simulated on each frame's grid with matched measurement noise, over the measured background S_q, against pixel size (30 frames). (b) Cross-scale RMSE of each measured zoom resampled through the engine onto its parent frame (16 pairs, child→parent scan numbers), with the parent's noise level. These values use the RMSE report's convention (one constant offset); the driver convention with a fitted plane gives the slightly lower ranges quoted in the text.

5.2. What these data cannot validate

The same measurements bound the claims that can be made:

1. *Tip geometry at the nominal scale* ($R \lesssim 12$ nm). No folder resolves it: grating walls are shallower than every facet, the polymer and fused-silica zooms give non-rejections without power, and PS-LDPE is pixel-limited. Only blunt apexes are excluded.
2. *Yaw sign and cantilever axis*, which centrosymmetric pits cannot reveal.
3. *Absolute XY and Z calibration.* No traceable standard was available; the scale factors found are session values.
4. *Feedback dynamics, holder tilt, mechanics and force-dependent indentation*, which are outside the model.

A benchmark that reports agreement on these samples should therefore not be read as evidence that the simulator's tip models are correct at the nanometre scale.

5.3. Implications for simulator-based inverse methods

Three practical points follow for anyone who trains or tests reconstruction methods on simulated data.

1. *Header metadata is not ground truth.* The nominal probe radius in the file header was contradicted twice: by a factor of about 65 for the sphere probe, and for the 125 nm RTESPA apex relative to its cross-probe partner. Folder labels also conflict with headers ("SNL10 B" against an RTESPA-300-like header). Independent probe characterisation (SEM, certificates, tip-check samples) is a prerequisite, not an option.
2. *Pixel response belongs in the forward model.* The instrument behaves as if each pixel were averaged over about one pixel along the fast axis. Ignoring this inflates an inferred radius about threefold, from 153 to 476 nm on CS-20NG.
3. *Samples must be tip-sensitive at the scale of interest.* Micrometre gratings with shallow walls validate sample models and registration, not tips. A validation of tip effects needs features sharper than the probe at the nanometre scale, imaged at pixel sizes comparable to the apex radius.

5.4. Limitations

Each grating folder has only 1–3 held-out frames at 1–3 independent sites; one VGRP-GS hold-out probably re-images the calibrate pits, and the SNL10 B hold-out is one field imaged twice. The in-folder tip discrimination is a heuristic ratio, not a statistical test, and the cross-probe rule cannot reach "inconsistent" at the available site counts. Quoted uncertainties are within-session repeatability. Widths and corner radii depend on the estimator by 10–30 nm. Probe identities are unverified: there was no SEM image, certificate or thermal tune.

6. Conclusions

We validated an open, geometric AFM tip-convolution simulator against 51 real Bruker frames from ten folders, using held-out frames, candidate-probe sweeps, a no-tip control, evidence tags and independent verification of every headline claim. A measured grating model transferred to held-out sites reproduces the images to 2.8–6.3 nm RMS, 5–18 times lower than the nominal presets. The same data reject the probe radius stated in the file headers and show that the fast-axis pixel response must be modelled. They do not identify the tip at the nanometre scale, the yaw, or the absolute calibration. These limits are the boundary conditions for the next step, the inverse problem:

recovering known calibration geometry from real scans, with reconstruction quality measured against independently characterised probes.

Code and data availability

The simulator (browser applications and Python engine), the validation pipeline, the case catalogue and the tests are available at [TODO: repository URL / DOI]. The raw Bruker data are [TODO: available from the authors on request / deposited at …].

CRediT authorship contribution statement

[TODO: Fill in per author: Conceptualization, Software, Validation, Formal analysis, Writing.]

Declaration of competing interest

[TODO: The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.]

Acknowledgements

[TODO: Instrument operators, facility, funding.]

References

[1] G. Binnig, C. F. Quate, C. Gerber, Atomic force microscope, Physical Review Letters 56 (1986) 930–933. doi:10.1103/PhysRevLett.56.930.

[2] D. Keller, Reconstruction of STM and AFM images distorted by finite-size tips, Surface Science 253 (1991) 353–364. doi:10.1016/0039-6028(91)90606-S.

[3] J. S. Villarrubia, Algorithms for scanned probe microscope image simulation, surface reconstruction, and tip estimation, Journal of Research of the National Institute of Standards and Technology 102 (1997) 425–454. doi:10.6028/jres.102.030.

[4] D. Nečas, P. Klapetek, Gwyddion: an open-source software for SPM data analysis, Central European Journal of Physics 10 (2012) 181–188. doi:10.2478/s11534-011-0096-2.

[5] Gwyddion developers, Gwyddion user guide: Tip convolution artefacts, http://gwyddion.net/documentation/user-guide-en/tip-convolution-artefacts.html, accessed 2026.

[6] V. Kocur, et al., Correction of AFM data artifacts using a convolutional neural network trained with synthetically generated data, Ultramicroscopy 246 (2023) 113666. doi:10.1016/j.ultramic.2022.113666.

[7] L. K. S. Bonagiri, et al., Precise surface profiling at the nanoscale enabled by deep learning, Nano Letters (2024). doi:10.1021/acs.nanolett.3c04712.

[8] Y. Matsunaga, et al., End-to-end differentiable blind tip reconstruction for noisy atomic force microscopy images, Scientific Reports 13 (2023) 129. doi:10.1038/s41598-022-27057-2.

[9] R. Millan-Solsona, et al., Synthetic data-driven deep learning for label-free autonomous atomic force microscopy, Nature Communications 17 (2026) 3886. doi:10.1038/s41467-026-70421-3.

[10] S. Chen, et al., Multi-view neural 3D reconstruction of micro- and nanostructures with atomic force microscopy, Communications Engineering 3 (2024) 131. doi:10.1038/s44172-024-00270-9.

[11] J. Dai, et al., Accurate tip characterization in critical dimension atomic force microscopy, Measurement Science and Technology 31 (2020) 085013. doi:10.1088/1361-6501/ab7fd2.

[12] TODO: add authors, A novel method for determining AFM probe tip characteristics using trench-based reference material, Nanomanufacturing and Metrology (2025). doi:10.1007/s41871-025-00262-z.

[13] N. Otsu, A threshold selection method from gray-level histograms, IEEE Transactions on Systems, Man, and Cybernetics 9 (1979) 62–66. doi:10.1109/TSMC.1979.4310076.